

The Exact Boundary of Period Inflation: A Machine-Checked Characterization for the Winnower CA Background Detector

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Abstract

Winnower is an open-source tool that splits a cellular-automaton spacetime into a periodic *background* and a *residual mask*, scanning candidate periodicities and choosing among them with a complexity-aware score. Like every fit-based detector it faces *period inflation*: a longer-period candidate can never fit worse, so an unpenalized selector drifts toward the largest model. We report a result that pins down exactly when this pressure exists. For two constant-velocity candidates, five natural conditions — velocity-matched divisibility, orbit-partition refinement, model-class nesting, universal residual improvement, and universal likelihood improvement — are all *equivalent*, and divisibility is *necessary*, not merely sufficient: whenever it fails, some spacetime makes the longer candidate fit strictly worse. The full equivalence, including the previously open necessity direction, is now machine-checked in Lean 4 under public continuous integration, and an exhaustive finite-window test confirms it over every candidate pair on a small grid. The characterization is the precise justification for Winnower’s complexity penalty, and it is reproducible and inspectable in the tool’s live web companion.

Motivation

Cellular automata (CA) routinely produce spacetimes in which broad periodic domains coexist with defects and particle-like activity; those domains are the canvas against which coherent structures such as gliders become visible (Hanson and Crutchfield, 1992; Crutchfield and Hanson, 1997). *Winnower* is a fast global front end for this setting: given a binary spacetime U and a grid of *candidates* (p, s) — “the pattern repeats every p steps, drifting sideways by s ” — it groups the cells each candidate forces to be equal (its *orbit classes*), takes a majority vote inside each class to build the best-fitting background B^* , and records the residual mask $M = U \oplus B^*$. Each candidate costs one linear pass, so whole-catalog 1D/2D/3D surveys are practical, and every reported number re-runs in an in-browser demo.

A selector that scores candidates by fit alone has a built-in bias. Call (p_2, s_2) a *velocity-matched multiple* of (p_1, s_1) when $p_2 = m p_1$ and $s_2 = m s_1 \bmod \mathbf{D}$: the same drift expressed with a longer period. Such a multiple splits every orbit class into finer pieces, so its optimal residual count

and its negative log-likelihood (NLL) can only improve. Unpenalized selection therefore inflates the period whether or not any extra structure is present. Winnower answers this with a complexity term — a Bernoulli normalized-maximum-likelihood (NML) penalty (Shtarkov, 1987; Grünwald, 2007) — but a penalty is only well-motivated if we know *exactly* which comparisons carry the inflation pressure. This abstract reports that exact boundary and its machine-checked proof.

An Exact Characterization

Fix a torus with spatial dimensions $\mathbf{D}$ and temporal length T , and two candidates c_1, c_2 acting by spacetime translation. Write Π_i for the orbit partition of c_i , $\mathcal{B}_i$ for the model class of fields constant on Π_i -classes, $d_U^*(c_i)$ for the minimum Hamming distance from U to $\mathcal{B}_i$, and $\text{NLL}_U(c_i)$ for the orbit-class Bernoulli negative log-likelihood.

Theorem (equivalence). The following are equivalent:

- (i) c_2 is a velocity-matched multiple of c_1 ;
- (ii) c_2 ’s translation is a power of c_1 ’s;
- (iii) Π_2 refines Π_1 ;
- (iv) $\mathcal{B}_1 \subseteq \mathcal{B}_2$;
- (v) $d_U^*(c_2) \leq d_U^*(c_1)$ for *every* U ;
- (vi) $\text{NLL}_U(c_2) \leq \text{NLL}_U(c_1)$ for *every* U .

The forward chain (i) $\Rightarrow$ $\dots$ $\Rightarrow$ (vi) is routine (cyclic-group structure; Jensen’s inequality for the concavity of binary entropy). The content is the *necessity* direction: if divisibility fails, refinement fails, and then a concrete adversarial spacetime — constant on Π_1 -classes but mixing two of them inside one Π_2 -class — makes c_2 strictly worse under both d^* and NLL. Closing the loop requires showing that partition refinement *forces* a power (iii) $\Rightarrow$ (ii): on the torus, orbits are cosets of the cyclic subgroup a translation generates, so refinement at the identity yields subgroup containment, and a translation is fixed by its value at one point. Divisibility is thus the exact frontier of the guaranteed-improvement regime: on one side longer periods always win; on the other, the data can always punish them. This is precisely the set of comparisons Winnower’s penalty must, and does, neutralize.

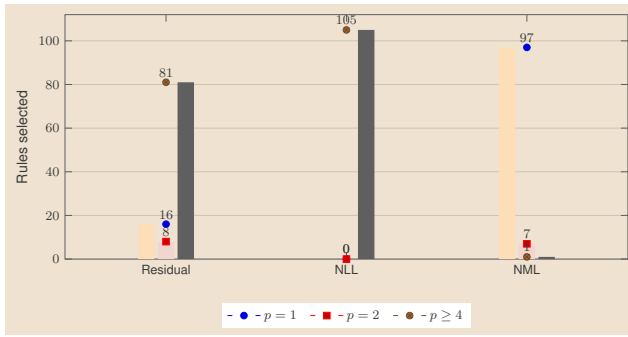


Figure 1: Why the boundary matters in practice. On the 105 non-degenerate Life-like rules at horizon $T=100$, selecting by fit alone (raw residual count, or plain NLL) is pulled to the largest scanned period — the inflation the theorem predicts along every velocity-matched chain — while the complexity-penalized score (Bernoulli NML) concentrates at low period. Plain NLL selects the maximum period for *all* 105 rules.

Machine-Checked, and Reproducible in the Tool

We regard a mathematical claim as verified only when Lean’s kernel accepts it. The full six-way equivalence, including the necessity direction (iii) $\Rightarrow$ (ii) that earlier drafts left open, is proved in Lean 4 against a pinned Mathlib toolchain and rebuilt on every push by the project’s public continuous-integration workflow, which additionally greps the built artifact for placeholder tactics. Independently, an exhaustive test enumerates all candidate pairs on a 6×4 space-time window and checks that partition refinement holds *iff* velocity-matched divisibility holds — a finite-window counterpart to the torus theorem, and a guard against silent formalization drift.

The characterization is not a footnote to the tool; it is its justification, and it is inspectable where the tool lives. The Winnower web companion carries a dedicated “Late-Breaking” section stating the theorem in plain terms and linking the machine-checked Lean file and its public CI runs; and the same in-browser reproduction that backs every other number reports, for any supported rule, seed, and horizon, the per-period fit (NLL) and penalized (NML) scores side by side — so a visitor can watch the fit column improve monotonically along velocity-matched chains, exactly as the theorem dictates, while the penalized column resists.

Status and Next Steps

This is work in progress. The empirical picture (planted-ground-truth recovery at 1.00 joint accuracy over 642 runs; zero false positives on shuffled and random controls; the ablation above) is established in the companion full paper; the new contribution here is the exact, machine-checked characterization that explains *why* the penalty is the right shape and

exactly where it is needed. Open threads include lifting the finite-window necessity direction (which currently carries one nondegeneracy hypothesis, $p_2 \leq T - 1$) into Lean alongside the torus form, and extending the analysis from pairwise comparisons to the full candidate lattice the scanner actually searches. The tool, the proofs, the tests, and the in-browser reproduction are public.

References

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